

1) Configuring a Basic Spring Application

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>

    <dependencies>
        <!-- Spring Core -->
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
            <version>6.1.8</version>
        </dependency>
    </dependencies>

</project>
```

Repository Class

```
package com.library.repository;
```

```
public class BookRepository {  
  
    public void displayBook() {  
        System.out.println("Book data fetched from repository");  
    }  
}
```

Service Class

```
package com.library.service;  
  
import com.library.repository.BookRepository;  
  
public class BookService {  
  
    private BookRepository bookRepository;  
  
    public void setBookRepository(BookRepository bookRepository) {  
        this.bookRepository = bookRepository;  
    }  
  
    public void showBook() {  
        System.out.println("Book Service Called");  
        bookRepository.displayBook();  
    }  
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <!-- Repository Bean -->
  <bean id="bookRepository"
    class="com.library.repository.BookRepository"/>

  <!-- Service Bean -->
  <bean id="bookService"
    class="com.library.service.BookService">

    <property name="bookRepository"
      ref="bookRepository"/>

  </bean>

</beans>
```

Main Class

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
public class MainApp {  
  
    public static void main(String[] args) {  
  
        ApplicationContext context =  
            new ClassPathXmlApplicationContext("applicationContext.xml");  
  
        BookService service =  
            (BookService) context.getBean("bookService");  
  
        service.showBook();  
    }  
}
```

2) Implementing Dependency Injection

BookRepository.java

```
package com.library.repository;  
  
public class BookRepository {  
  
    public void displayBook() {  
        System.out.println("Fetching book details from repository...");  
    }  
}
```

BookService.java

```
package com.library.service;
```

```
import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    // Setter Injection
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showBook() {
        System.out.println("BookService is executing...");
        bookRepository.displayBook();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="
            http://www.springframework.org/schema/beans
            https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Repository Bean -->
```

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>
```

```
<!-- Service Bean -->
```

```
<bean id="bookService"
      class="com.library.service.BookService">
```

```
    <property name="bookRepository"
              ref="bookRepository"/>
```

```
</bean>
```

```
</beans>
```

Main Class

```
package com.library;
```

```
import com.library.service.BookService;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
public class LibraryManagementApplication {
```

```
    public static void main(String[] args) {
```

```
        ApplicationContext context =
```

```
            new ClassPathXmlApplicationContext("applicationContext.xml");
```

```
BookService service =  
    (BookService) context.getBean("bookService");  
  
service.showBook();  
}  
}
```

3) Creating and Configuring a Maven Project

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="  
        http://maven.apache.org/POM/4.0.0  
        http://maven.apache.org/xsd/maven-4.0.0.xsd">  
  
    <modelVersion>4.0.0</modelVersion>  
  
    <groupId>com.library</groupId>  
    <artifactId>LibraryManagement</artifactId>  
    <version>1.0-SNAPSHOT</version>  
  
    <dependencies>  
  
        <!-- Spring Context -->  
  
        <dependency>  
            <groupId>org.springframework</groupId>  
            <artifactId>spring-context</artifactId>  
            <version>5.3.30</version>
```

</dependency>

<!-- Spring AOP -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-aop</artifactId>

<version>5.3.30</version>

</dependency>

<!-- Spring Web MVC -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-webmvc</artifactId>

<version>5.3.30</version>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.11.0</version>

<configuration>

<source>1.8</source>


```
        <target>1.8</target>
    </configuration>
```

```
</plugin>
```

```
</plugins>
```

```
</build>
```

```
</project>
```

1) Configuring a Basic Spring Application

Expected Output

Book Service Called

Book data fetched from repository

2) Implementing Dependency Injection

Expected Output

BookService is executing...

Fetching book details from repository...

3) Creating and Configuring a Maven Project

This exercise focuses on configuring a Maven project using pom.xml. It does not produce any console output.

Expected Output

BUILD SUCCESS

If the project is compiled using Maven, you should see a successful build message similar to:

```
[INFO] -----
```

```
[INFO] BUILD SUCCESS
```

```
[INFO] -----
```

[INFO] Total time: 2.3 s

[INFO] -----